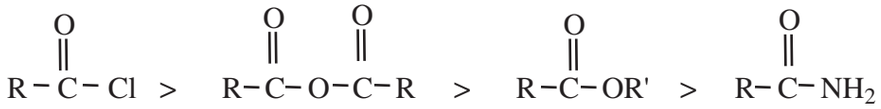
Relative reactivity of Acid derivatives

The reactivity of the acid derivatives follows the order



The above order of reactivity can be explained in terms of

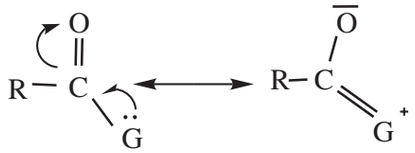
i) Basicity of the leaving group ii) Resonance effect

(i) Basicity of the leaving group

Weaker bases are good leaving groups. Hence acyl derivatives with weaker bases as leaving groups (L) can easily rupture the bond and are more reactive. The correct order of the basicity of the leaving group is $H_2\bar{N}$: > : $\bar{O}R$ > $RCO\bar{O}$: > : $\bar{Cl}$ Hence the reverse is the order of reactivity.

(ii) Resonance effect

Lesser the electronegativity of the group, greater would be the resonance stabilization as shown below.

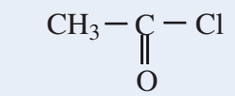
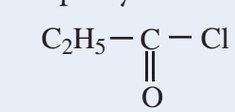
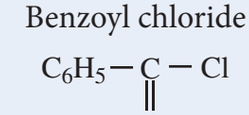


This effect makes the molecule more stable and reduces the reactivity of the acyl compound. The order of electronegativity of the leaving groups follows the order – Cl > - OCOR > - OR > - NH_2

Hence the order of reactivity of the acid derivatives with nucleophilic reagent follows the order

acid halide > acid anhydride > esters > acid amides

12.14.1 Nomenclature

Compound (common name, Structural formula, IUPAC Name)	IUPAC Name			
	Prefix with position number	Root used	Primary suffix	Secondary Suffix
Acetyl chloride  Ethanoylchloride	–	eth	ane	oyl chloride
Propionyl chloride  Propanoylchloride	–	prop	ane	oyl chloride
Benzoyl chloride  Benzoylchloride	–	Benz	ane	oyl chloride

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